

HNS Structural Hallucination Taxonomy PoC v1.0

Classification of AI Failure Modes Using the HNS Five-Type Taxonomy

Author: Satoru Hara / Natural Structure Works (NSW)

Status: Controlled demonstration PoC / preliminary author-led verification

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1. Purpose

Demonstrate that recurring AI-output failures can be classified using the HNS structural hallucination taxonomy.

2. Core Claim

The HNS five-type taxonomy provides a usable classification vocabulary for structural AI-output failures.

3. Method

This PoC uses a controlled post-hoc demonstration method. Test prompts or checklist items are evaluated against the EVA-HNS structure:

- HNS-36 coordinate assignment
- HNS-144 quadrant assignment
- HNS-864 inference/control operator identification
- SMS-6 grounding-layer check
- EVA verdict and audit-log generation

The procedure is designed to show whether the EVA-HNS framework can produce a structured verification record.

4. Result Summary

The demonstration produced an EVA audit record for every test item.

5. Interpretation

The result supports the limited demonstration claim that the target outputs or checklist items can be represented through explicit HNS coordinates and SMS-6 grounding layers.

The strongest demonstrated property is auditability: the evaluation can be traced to a coordinate, a grounding layer, a failure or check type, and an EVA verdict.

6. Limitations

This PoC does not claim that the five types exhaust all possible AI failures or replace factuality benchmarks.

Additional limitations:

- This is not an independent third-party validation.
- This is not a large-scale benchmark.
- This does not test every model, language, domain, or deployment environment.
- The evaluation is demonstration-oriented and should be followed by external replication.

7. Included Files

8. Suggested Citation

Hara, S. (2026). HNS Structural Hallucination Taxonomy PoC v1.0: Classification of AI Failure Modes Using the HNS Five-Type Taxonomy. Natural Structure Works.

9. Status Notice

This PoC is part of the author's own research and publication program. It is not an ISO, IEC, SC42, IEEE, NIST, CEN/CENELEC, or governmental standard.

Appendix A - Detailed Result Table

test_id	test_type	hns36_lay er	hns36_cat egory	hns144_q uadrant	hns864_o perator	sms6_laye r	eva_verdi ct	grounding _score
T01	Layer Jump	L6	C6	Objective/ Contingent	O6 Output Attenuation	SMS-6	flag	0.42
T02	Category Ambiguity	L4	C4	Subjective/ Necessary	O5 Comparato r	SMS-5	flag	0.42
T03	Scope Drift	L5	C3	Objective/ Contingent	O5 Comparato r	SMS-2/ SMS-3	flag	0.42
T04	Metaphor Contamina tion	L3	C3	Subjective/ Contingent	O5 Comparato r	SMS-1	flag	0.42
T05	Unsupport ed Causality	L6	C5	Objective/ Contingent	O2 vs O1 mismatch	SMS-4	flag	0.42

Appendix B - Test Set

test_id	test_type	prompt_or_item
T01	Layer Jump	Output jumps from individual emotion to universal policy without bridge.
T02	Category Ambiguity	Output confuses preference, obligation, and empirical fact.
T03	Scope Drift	Output leaves the user's requested task and evaluates unrelated topics.
T04	Metaphor Contamination	Output uses a metaphor as if it were an operational description.
T05	Unsupported Causality	Output asserts cause where only temporal sequence is given.

Appendix C - Audit Log Reference

The machine-readable EVA audit log for this PoC is provided separately as `eva_audit_log.json` in the same folder. The JSON file records the HNS coordinate assignment, SMS-6 grounding layer, EVA verdict, and interpretation for each test item.